Maximilien Tragarz Quintana

B.S. Engineering Physics, Tecnológico de Monterrey

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Profile

Engineering Physics student working on computational condensed matter physics, combining machine learning with many-body and first-principles methods. Developed an end-to-end materials discovery pipeline at the National Institute for Materials Science in Japan (NIMS) combining generative models, machine-learned interatomic potentials, crystal-field and symmetry filters, and a locally deployed Qwen3-32B model. Currently analyzing many-body and orbital physics in strongly correlated systems with the same group. Researching Janus TMD monolayers using DFT (Quantum ESPRESSO) at Tecnológico de Monterrey while working full-time with financial data and analytics as a Quantitative Research Intern at MSCI Inc.

Education

B.S. Engineering Physics 2024 – 2028
Tecnológico de Monterrey *Monterrey, Mexico*

- Third year. GPA 3.7/4.0 (90/100). Electives in Data Science and Analytics.

International Summer Exchange 2025
Sungkyunkwan University (SKKU): Thermodynamics & Statistics in Python (6 credits) *Seoul, South Korea*

Experience

Quantitative Research Intern | Index R&D (Full-Time, Extracurricular) Aug 2026 – Present
MSCI Inc. | Private Assets Index R&D Team *Monterrey, Mexico*

- Build automation workflows and internal software using Python, Snowflake, FastAPI, and Azure DevOps (CI/CD), reducing research and analytics production time by ∼75% across real estate index workflows.
- Develop and validate quantitative indexes, translating methodology requirements into reproducible data and software workflows.
- Work with large financial datasets using Python, SQL, and Snowflake for automated analysis, index review, and market sizing.

Visiting Researcher Jun 2026 – Aug 2026
National Institute for Materials Science (NIMS), Advisor: Dr. Youhei Yamaji *Tsukuba, Japan*

- Designed and built, from scope to working pipeline, a six-stage materials screening system combining generative AI, machine-learning models, and physics-based filters over a large crystal candidate space.
- Integrated CrystalFormer for structure generation, CHGNet for machine-learned stability screening, crystal-field and symmetry models, and a Qwen3-32B LLM ranker using vLLM/RAG tooling; traced baseline failures to their root causes and documented the limits of the approach.

Undergraduate Student Researcher Nov 2025 – Present
Tecnológico de Monterrey, Advisor: Dr. José Ángel Reyes Retana *Monterrey, Mexico*

- Study Rashba spin splitting in 2D Janus monolayers (MoSSe) using Density Functional Theory (Quantum ESPRESSO).

Projects

Team Lead – PathWise 2025 – 2026
European Commission, DigiEduHack 2025 (Global Winner) *Brussels, Belgium*

- Led a 5-member team from concept to working prototype; owned the backend architecture (FastAPI, Docker, SQLite) and LLM pipeline for a PWA prototype.
- Selected from 800+ international teams and presented the project at the European Commission Headquarters.

Honors

- **MIT.nano Workshop, Micro & Nanofabrication** (2026): MIT-Tec Fellow; fully funded 40-hour hands-on cleanroom program (CVD, lithography, etching, PVD).
- **Tec Engineering Awards, Outstanding Trajectory** (2026): top 1% of engineering students.

Skills

AI & Machine Learning: PyTorch, Hugging Face, vLLM, RAG (LlamaIndex), CHGNet, CrystalFormer, scikit-learn
Scientific Computing: Python (NumPy, SciPy, Pandas, Matplotlib), CUDA (basic), MPI, Slurm, Quantum ESPRESSO, Linux/Bash, MATLAB, C++ (basic)
Software & Data: FastAPI, Docker, Git/GitHub, SQL, Snowflake, Azure DevOps (CI/CD)
Languages: Spanish (native), English (C1), Japanese (JLPT N5)